# MODEL PERFORMANCE ANALYSIS

*Regression Task — Validation & Test Metrics*
*ITAI 1371 | Spring 2026*

| Model | Validation Set | | | Test Set | | |
|---|---|---|---|---|---|---|
| | Val_MAE | Val_MSE | Val_R2 | Test_MAE | Test_MSE | Test_R2 |
| Linear Regression | 0.3073 | 0.183 | 0.79 | 0.3077 | 0.1853 | 0.7553 |
| Decision Tree Regressor | 0.3834 | 0.3127 | 0.6412 | 0.3707 | 0.2863 | 0.6218 |
| Random Forest Regressor | 0.3126 | 0.1823 | 0.7908 | 0.2992 | 0.1786 | 0.7641 |
| Gradient Boosting Regressor | 0.2956 | 0.1676 | 0.8077 | 0.2842 | 0.1656 | 0.7813 |
| K-Nearest Neighbors Regressor | 0.3096 | 0.1788 | 0.7948 | 0.305 | 0.1804 | 0.7617 |
| Voting Ensemble | 0.2924 | 0.1615 | 0.8147 | 0.283 | 0.1601 | 0.7885 |
| Bayesian Ensemble | 0.2913 | 0.1582 | 0.8184 | 0.2812 | 0.1594 | 0.7894 |

# Which model performed best and worst?

## Best Model: Bayesian Ensemble

The Bayesian Ensemble achieved the highest $R^2$ scores and the lowest MAE and MSE values across the validation and test datasets. By combining predictions from multiple strong models, it improved overall prediction accuracy and produced more stable results compared to individual regression models.

## Worst Model: Decision Tree Regressor

The Decision Tree Regressor showed the lowest $R^2$ values and higher prediction errors compared to other models. Its weaker performance suggests overfitting and poor generalization on the test data

## Conclusion

The regression models were successfully evaluated using validation and test metrics. The Bayesian Ensemble model achieved the best performance, while the Decision Tree Regressor showed the weakest results. Ensemble learning improved overall prediction accuracy and produced more reliable results compared to individual models.